



Explorers and raiders
The Vikings of Scandinavia were the most intrepid sailors of medieval Europe, braving the Atlantic to reach Greenland and North America. The horn (*left*) was used for signalling between ships at night and in foggy weather. The settled populations of Europe lived in constant dread of their destructive raids.

LIMITED WAR

Through the rise and fall of Mediterranean naval powers, there were inevitably changes in galley construction, tactics, and weaponry. But what was most remarkable was perhaps the conservatism of combat between galleys: advance to battle in line abreast; a preliminary exchange of missiles; ramming or grappling, followed by boarding of enemy ships. Such tactics served for 2,000 years. The limitations of galley warfare were considerable. Relatively small ships with a large complement of oarsmen and marines, galleys could not carry supplies to stay at sea for periods longer than a few days. Nor were they suited to sailing in heavy seas. Galley warfare was thus restricted to coastal waters, and almost always ceased in the winter months when the weather was too rough. These limitations did not matter much until the rise of the sailing warship in northern and western Europe presented a direct challenge to the reign of the war galley and to the central importance of the Mediterranean as a theater of naval warfare.

NORTHERN WATERS

Galleys were never confined exclusively to the Mediterranean. The Viking longships that terrorized northern Europe in the 9th century, with their mix of oars and sails, were a kind of galley. Oared ships played an important part in warfare in the Baltic into the 18th century. In medieval times merchant galleys sailed in convoys from the Mediterranean to England and Flanders, and war galleys were used by the kings of France and Castile in battles against the English kings in the English Channel and the Bay of Biscay. But countries with Atlantic shorelines had their own shipbuilding and ship-fighting traditions more suited to rougher, wider waters. The round ships known as cogs, developed initially as merchant vessels, became warships, because navies took over merchant ships in times of war. They provided a high platform for bowmen and men-at-arms in battles such as Sluys, fought between the English and French in 1340—essentially a land battle fought on water. By the end of the 15th century, ships relying on oars for even part of their propulsion had only a subsidiary role to play in warfare in most northern waters, supporting larger sailing ships such as carracks and caravels.

ASIAN DEVELOPMENTS

Asia and the Indian Ocean, meanwhile, had their own traditions of naval warfare. Most of this took place in coastal or inland waters and was a direct adjunct to land warfare. By the time a permanent Chinese navy was founded by the Song dynasty in 1132, China had an array of diverse vessels including paddlewheel ships, galleys, and sailing ships. Exploiting the resources of a prosperous and populous state, China became the world's greatest naval power, although Europeans knew little or nothing about it. In the early 15th century the Ming dynasty embarked on naval power-projection on a vast scale with the voyages of Admiral Zheng He, who took a fleet of massive war junks around southeast Asia and across the



1210 First recorded sea battle, between the Hittites and the Cypriots

800 Foundation of Carthage by the Phoenicians, the leading maritime power in the Mediterranean

500 Greek city states such as Corinth and Athens develop the trireme, powered by three banks of oars and designed for ramming

Corinthian helmet One of Athens' chief rivals as a sea power, Corinth fought with the Spartans in the Peloponnesian War.

31 Battle of Actium: Octavian's defeat of Antony and Cleopatra gives Rome mastery of the Mediterranean

1200 BCE | 900 BCE | 600 BCE | 300 BCE | 1 CE

1190 Egyptians win sea battle against the "Sea Peoples"

Stone anchors Simple anchors consisting of stones with holes drilled in them have been found in all parts of the Mediterranean.



480 Battle of Salamis: decisive naval victory of the Greeks over the invading Persians

431–404 Peloponnesian War: Athens defeated by alliance of rival Greek city-states led by Sparta

264–241 First Punic War: Romans build fleet to challenge the Carthaginians for control of the western Mediterranean